

Rhianna Ruggiero DSC 630 Assignment Week 3 **Using Data to Improve MLB Attendance**

```
In [145... import pandas as pd
df = pd.read_csv("dodgers-2022.csv")
df.head()
```

```
Out[145... month day attend day_of_week opponent temp skies day_night cap shirt fire
0 APR 10 56000 Tuesday Pirates 67 Clear Day NO NO
1 APR 11 29729 Wednesday Pirates 58 Cloudy Night NO NO
2 APR 12 28328 Thursday Pirates 57 Cloudy Night NO NO
3 APR 13 31601 Friday Padres 54 Cloudy Night NO NO
4 APR 14 46549 Saturday Padres 57 Cloudy Night NO NO
```



```
In [147... df.tail()
```

```
Out[147... month day attend day_of_week opponent temp skies day_night cap shirt fir
76 SEP 29 40724 Saturday Rockies 84 Cloudy Night NO NO
77 SEP 30 35607 Sunday Rockies 95 Clear Day NO NO
78 OCT 1 33624 Monday Giants 86 Clear Night NO NO
79 OCT 2 42473 Tuesday Giants 83 Clear Night NO NO
80 OCT 3 34014 Wednesday Giants 82 Cloudy Night NO NO
```



Use this data to make a recommendation to management on how to improve attendance. Tell a story with your analysis and clearly explain the steps you take to arrive at your conclusion. This is an open-ended question, and there is no one right answer. You are welcome to do additional research and/or use domain knowledge to assist your analysis, but clearly state any assumptions you make.

First I will want to check for any missing data and to clean up the information before I analyze it.

```
In [245... df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 81 entries, 0 to 80
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   month                 81 non-null    object
1   day                   81 non-null    int64
2   attend                81 non-null    int64
3   day_of_week           81 non-null    object
4   opponent              81 non-null    object
5   temp                  81 non-null    int64
6   skies                 81 non-null    object
7   day_night             81 non-null    object
8   cap                   81 non-null    int64
9   shirt                 81 non-null    int64
10  fireworks              81 non-null    int64
11  bobblehead             81 non-null    int64
12  special_events         81 non-null    int64
dtypes: int64(8), object(5)
memory usage: 8.4+ KB
```

In [247...

df.describe()

Out[247...

	day	attend	temp	cap	shirt	fireworks	bobblehead	s
count	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000	81.000000	
mean	16.135802	41040.074074	73.148148	0.024691	0.037037	0.172840	0.135802	
std	9.605666	8297.539460	8.317318	0.156150	0.190029	0.380464	0.344713	
min	1.000000	24312.000000	54.000000	0.000000	0.000000	0.000000	0.000000	
25%	8.000000	34493.000000	67.000000	0.000000	0.000000	0.000000	0.000000	
50%	15.000000	40284.000000	73.000000	0.000000	0.000000	0.000000	0.000000	
75%	25.000000	46588.000000	79.000000	0.000000	0.000000	0.000000	0.000000	
max	31.000000	56000.000000	95.000000	1.000000	1.000000	1.000000	1.000000	

In [151...

df.isnull().sum()

Out[151...

month	0
day	0
attend	0
day_of_week	0
opponent	0
temp	0
skies	0
day_night	0
cap	0
shirt	0
fireworks	0
bobblehead	0
dtype: int64	

According to the above result there are no missing values in this data set.

In order to improve attendance, I will analyze the data to figure out which metric might influence the attend column. For example, if there are fireworks, will there be more attendance? Does a free bobblehead or shirt affect attendance or maybe the day of the week?

I will start by creating charts so that we can clearly see day of week vs number of people that attended the game. First I will look at the day of the week and group the attendance by it and see which day on average there is the highest attendance.

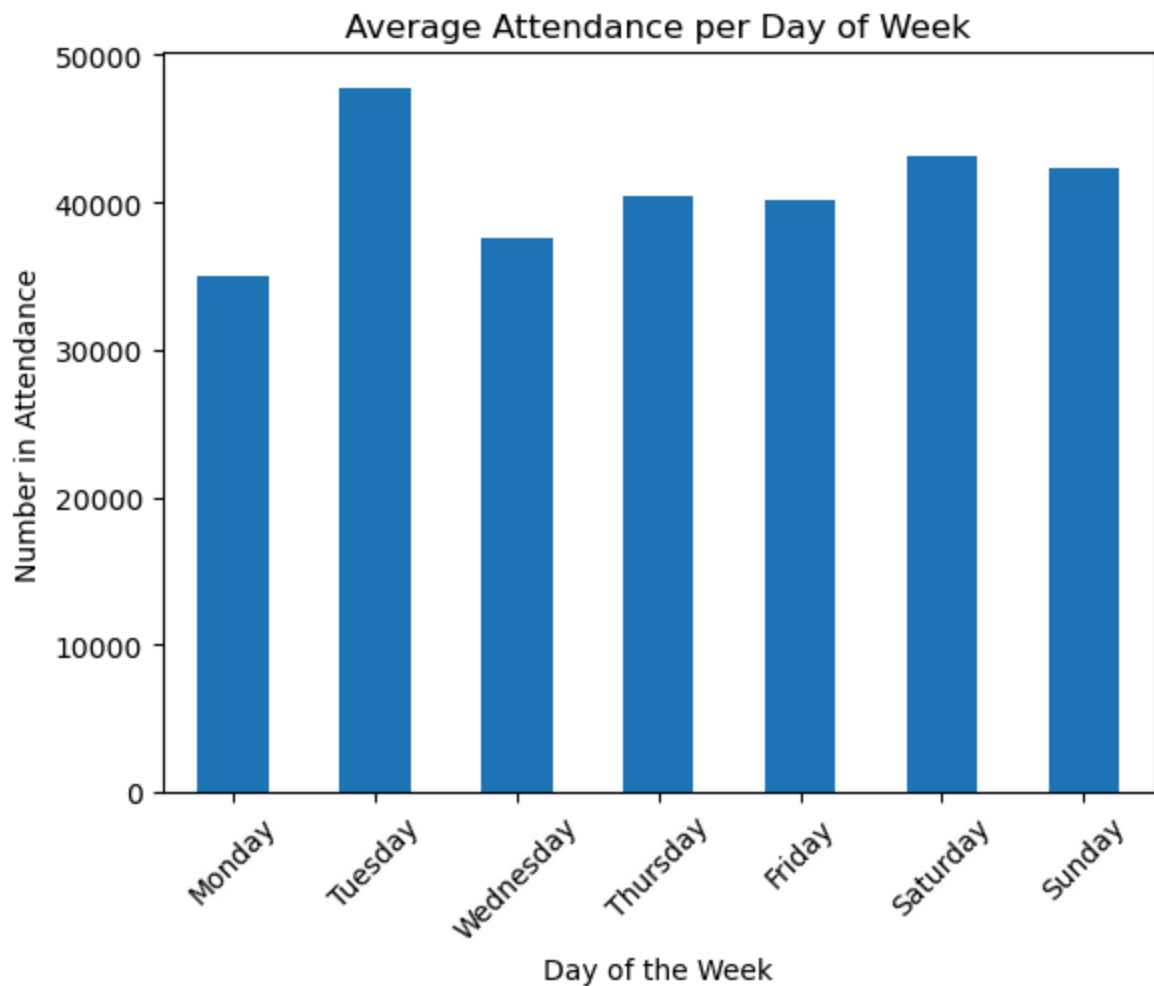
```
In [156... import matplotlib.pyplot as plt

dow_order = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "S

attend_dow = df.groupby("day_of_week")["attend"].mean().reindex(dow_order)
```

In order to have the resulting plot be in the correct weekday order, I had to create an ordered list and use it to reindex the dataframe that feeds the plot.

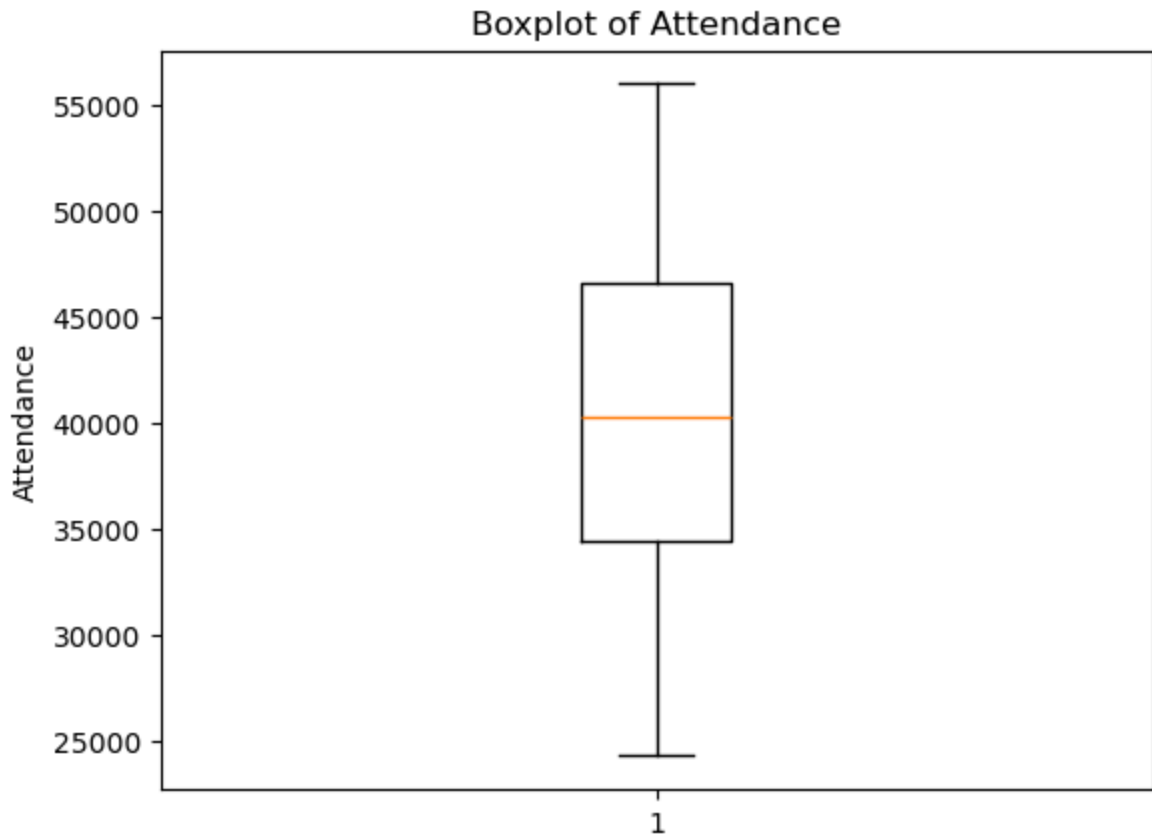
```
In [159... attend_dow.plot(kind="bar")
plt.title("Average Attendance per Day of Week")
plt.ylabel("Number in Attendance")
plt.xlabel("Day of the Week")
plt.xticks(rotation=45)
plt.show()
```



Based on this plot, Tuesdays have the highest attendance which I found surprising. Saturday and Sunday had the next highest and Monday had the lowest. I wonder if something is driving up attendance on Tuesdays like a free bobble head or T shirt or if there are any outliers causing this spike. I want to check for outliers for attendance and will use a boxplot to do so.

In [253...

```
plt.boxplot(df["attend"])
plt.title("Boxplot of Attendance")
plt.ylabel("Attendance")
plt.show()
```



This boxplot does not show any outliers (for boxplot/IQR standards) that would lead me to believe that something unique influenced higher attendance on Tuesdays.

Next I want to look at which days of the week have the most giveaways/special events including shirts, caps, fireworks and bobbleheads.

Since this data is listed as yes and nos, it would be easier to convert these to 1 for yes and 0 for no in order to group them together.

In [264...

```
specials = ["cap", "shirt", "fireworks", "bobblehead"]
df[specials] = df[specials].replace("YES",1).replace("NO",0)
df.head()
```

Out[264...

	month	day	attend	day_of_week	opponent	temp	skies	day_night	cap	shirt	fire
0	APR	10	56000	Tuesday	Pirates	67	Clear	Day	0	0	
1	APR	11	29729	Wednesday	Pirates	58	Cloudy	Night	0	0	
2	APR	12	28328	Thursday	Pirates	57	Cloudy	Night	0	0	
3	APR	13	31601	Friday	Padres	54	Cloudy	Night	0	0	
4	APR	14	46549	Saturday	Padres	57	Cloudy	Night	0	0	

Now I want to sum up all of the special events into one column and then group them together by week day to see which day has the most special events.

```
In [166...] df["special_events"] = df[["cap", "shirt", "fireworks", "bobblehead"]].sum(axis=1)
```

```
In [168...] events_by_dow = df.groupby("day_of_week")["special_events"].sum()
```

```
In [170...] print(events_by_dow)
```

```
day_of_week
Friday      13
Monday       1
Saturday     2
Sunday       3
Thursday     2
Tuesday      8
Wednesday    1
Name: special_events, dtype: int64
```

According to the above information, Fridays have had the most special events, however they don't have the most attendance compared to other days of the week. Tuesdays do have the next highest number of events though and had the highest attendance, so maybe the type of special event influences the attendance. Next I decided to look at specific special events and see which one occurs most on Tuesdays.

```
In [173...] caps_by_dow = df.groupby("day_of_week")["cap"].sum()
print(caps_by_dow)
```

```
day_of_week
Friday      0
Monday      0
Saturday    0
Sunday      1
Thursday    0
Tuesday     1
Wednesday   0
Name: cap, dtype: int64
```

```
In [175...] shirts_by_dow = df.groupby("day_of_week")["shirt"].sum()
print(shirts_by_dow)
```

```
day_of_week
Friday      0
Monday      1
Saturday    0
Sunday      1
Thursday    0
Tuesday     1
Wednesday   0
Name: shirt, dtype: int64
```

```
In [177...] fw_by_dow = df.groupby("day_of_week")["fireworks"].sum()
print(fw_by_dow)
```

```

day_of_week
Friday      13
Monday       0
Saturday     0
Sunday       0
Thursday     0
Tuesday      0
Wednesday    1
Name: fireworks, dtype: int64

```

```

In [179... bob_by_dow = df.groupby("day_of_week")["bobblehead"].sum()
print(bob_by_dow)

```

```

day_of_week
Friday       0
Monday       0
Saturday     2
Sunday       1
Thursday     2
Tuesday      6
Wednesday    0
Name: bobblehead, dtype: int64

```

According to the tables above, Tuesdays have had all of the events besides fireworks which mainly occurred on Fridays. No other events besides fireworks occurred on Fridays, which leads me to believe that fireworks might not be drawing crowds as much as other events, although there might be other factors in play like opponents or weather. Tuesdays have the highest attendance, and had one day of cap giveaways, one day of shirt giveaways and 6 days of bobblehead giveaways. Saturdays have the next highest attendance and had 2 days of bobblehead giveaways so I am going to look into the specific attendance on bobblehead giveaway days.

```

In [182... bob = df[df['bobblehead'] == 1]
shirt = df[df['shirt'] == 1]

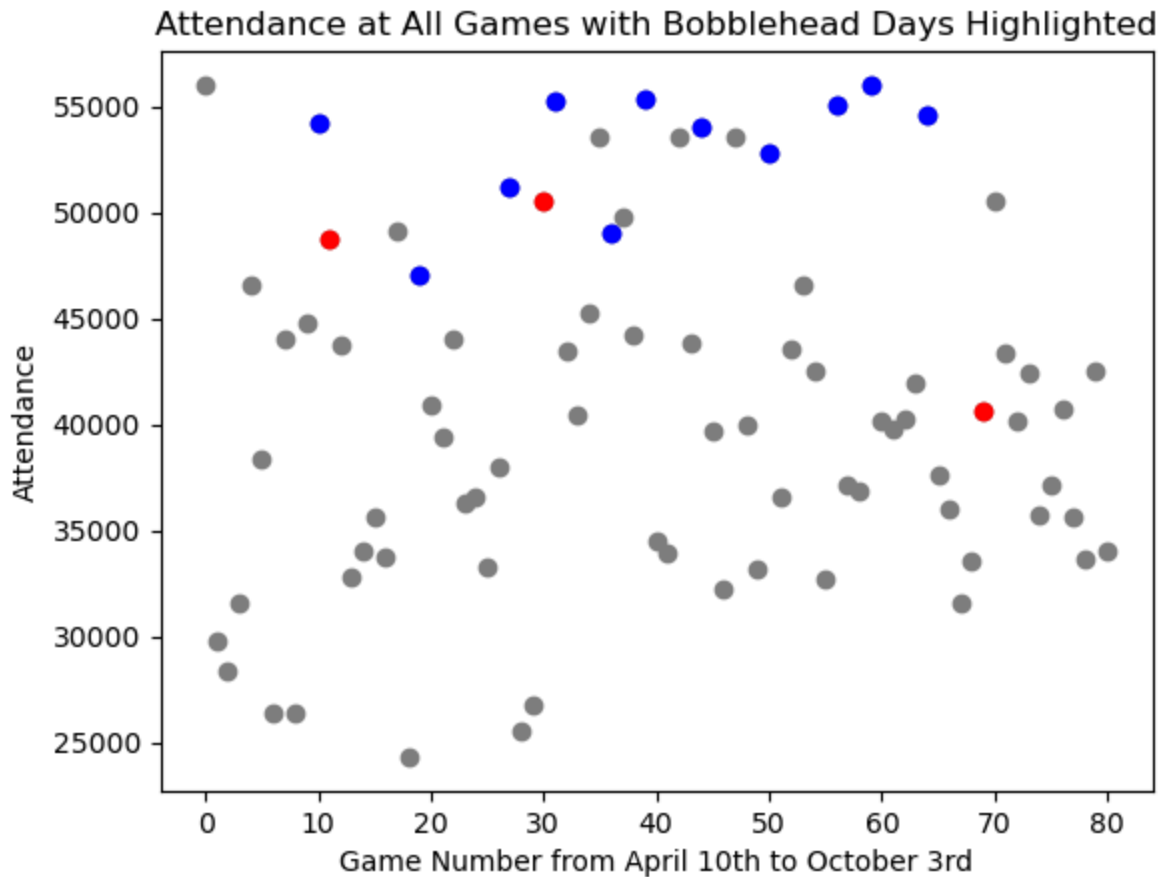
```

```

In [186... plt.scatter(df.index, df["attend"], color="gray")
plt.scatter(bob.index, bob["attend"], color="blue", label = "bobblehead")
plt.scatter(shirt.index, shirt["attend"], color="red", label = "shirt")

plt.title("Attendance at All Games with Bobblehead Days Highlighted")
plt.ylabel("Attendance")
plt.xlabel("Game Number from April 10th to October 3rd")
plt.show()

```



According to the chart above, days with bobblehead giveaways do have higher attendance in general from other days, however some other days without bobbleheads also have high attendance. I decided to add in shirts as well to see if they were given away on the highest attendance days as well. After doing this, shirts did not account for the other higher attendance days.

I was curious about the the top left data point with high attendance (but no bobblehead giveaway) and realized that this is the first game of the season which is bound to have high attendance since fans are excited that baseball is back and will want to go to the home opener game. With that information and thinking of the first game as an outlier, bobblehead days would be the games with the highest attendance overall. Next, I want to see if bobblehead giveaway days have the highest attendance compared to Tuesday games without bobbleheads or if some other factor might be influencing the data as well.

```
In [201...] tuesdays = df[df["day_of_week"] == "Tuesday"]
```

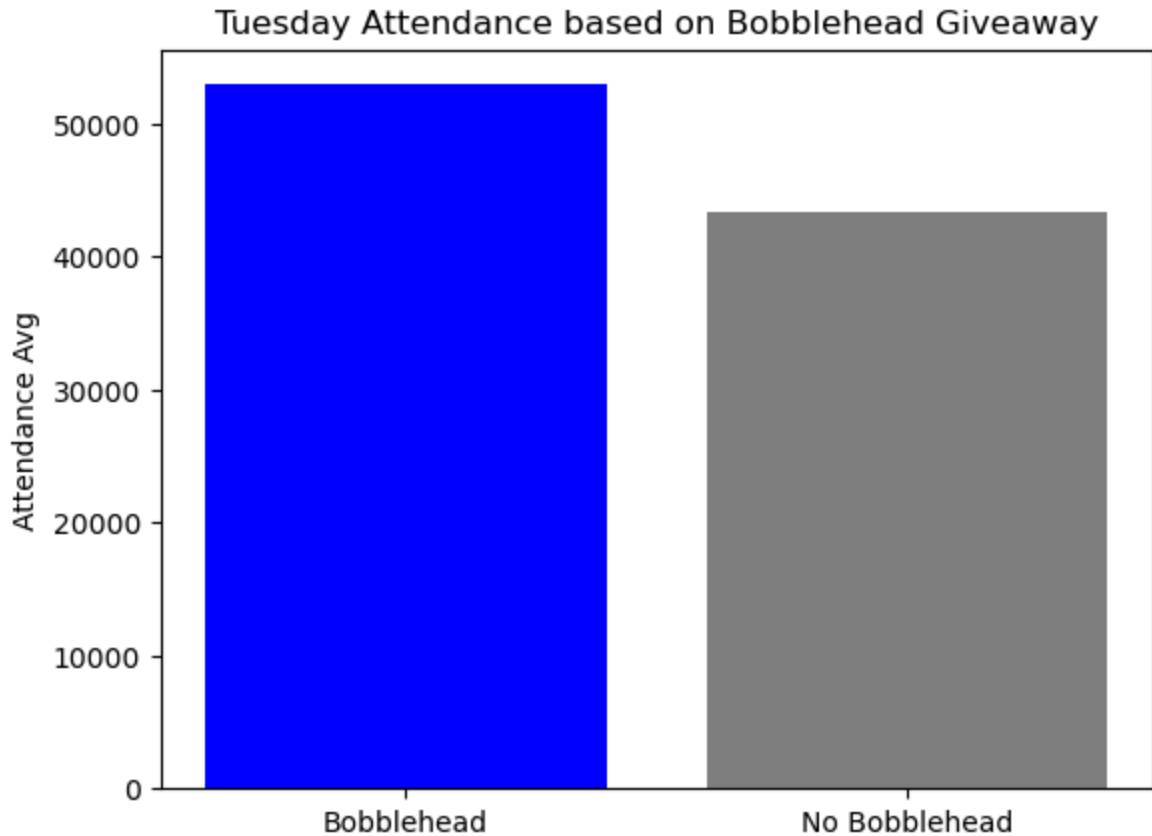
```
In [203...] tues_bob = tuesdays[tuesdays["bobblehead"] == 1]
tues_not_bob = tuesdays[tuesdays["bobblehead"] == 0]
```

```
In [308...] plt.bar(["Bobblehead"], [tues_bob["attend"].mean()], color="blue")
plt.bar(["No Bobblehead"], [tues_not_bob["attend"].mean()], color="gray")

plt.title("Tuesday Attendance based on Bobblehead Giveaway")
```



```
plt.ylabel("Attendance Avg")
plt.show()
```

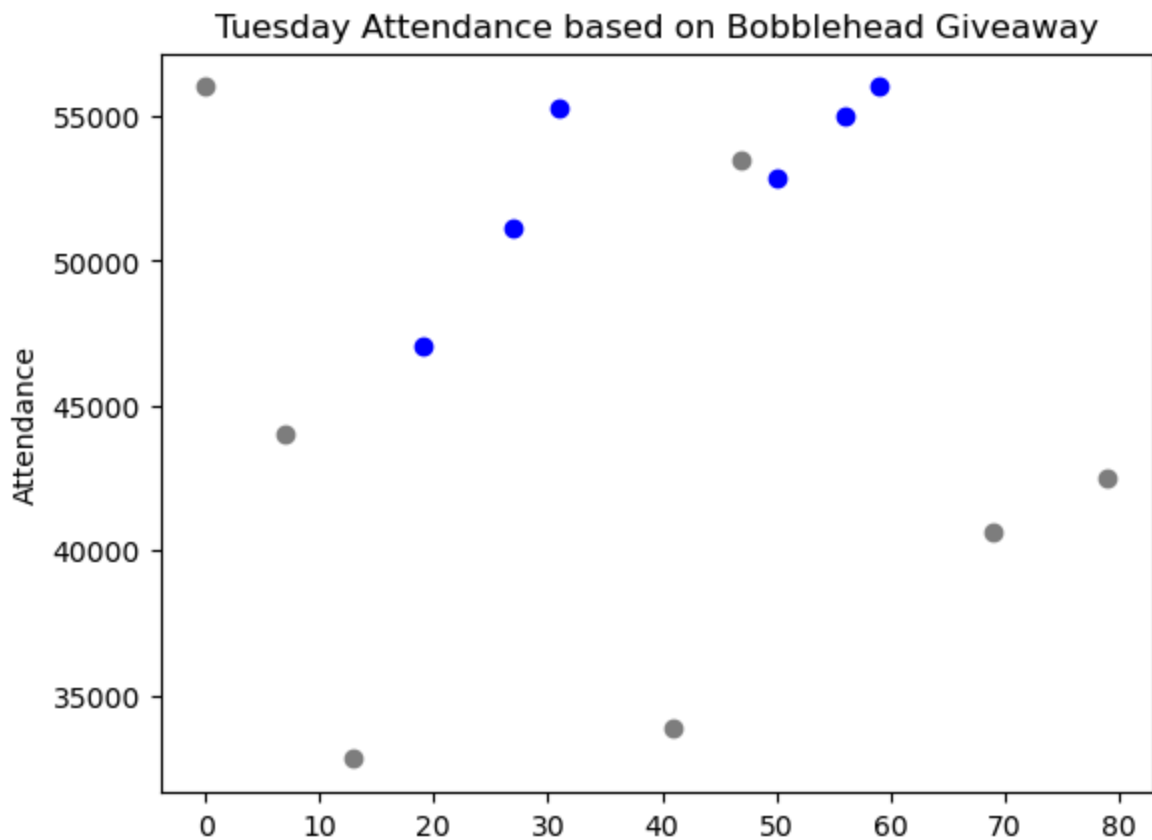


According to the chart above, Tuesdays with bobblehead giveaways do have higher attendance on average compared to Tuesdays without the bobble head giveaway. Next I would like to see attendance on Tuesdays in general as a scatterplot to see if one game's attendance is higher than the rest potentially skewing the average.

```
In [271... plt.scatter(tues_not_bob.index, tues_not_bob["attend"], color="gray")
plt.scatter(tues_bob.index, tues_bob["attend"], color="blue")

plt.title("Tuesday Attendance based on Bobblehead Giveaway")
plt.ylabel("Attendance")
```

```
Out[271... Text(0, 0.5, 'Attendance')
```



Opening day was on a Tuesday, so the first data point on the top left can be ignored and thought of as an outlier since the attendance most likely had to do with the game being on the first day of the season. Once that is ignored, you can see that bobblehead giveaway days have the most attendance compared to the other days, except for one game that still had less attendance than 3 other Tuesday games with bobbleheads. I'd like to see the specific data for this game to see what might have influenced the high attendance.

In [274...

```
tuesdays.head(20)
```

Out[274...

	month	day	attend	day_of_week	opponent	temp	skies	day_night	cap	shirt	fir
0	APR	10	56000	Tuesday	Pirates	67	Clear	Day	0	0	
7	APR	24	44014	Tuesday	Braves	63	Cloudy	Night	0	0	
13	MAY	8	32799	Tuesday	Giants	75	Clear	Night	0	0	
19	MAY	15	47077	Tuesday	Snakes	70	Clear	Night	0	0	
27	MAY	29	51137	Tuesday	Brewers	74	Clear	Night	0	0	
31	JUN	12	55279	Tuesday	Angels	66	Cloudy	Night	0	0	
41	JUL	3	33884	Tuesday	Reds	70	Cloudy	Night	1	0	
47	JUL	17	53498	Tuesday	Phillies	70	Clear	Night	0	0	
50	JUL	31	52832	Tuesday	Snakes	75	Cloudy	Night	0	0	
56	AUG	7	55024	Tuesday	Rockies	80	Clear	Night	0	0	
59	AUG	21	56000	Tuesday	Giants	75	Clear	Night	0	0	
69	SEP	4	40619	Tuesday	Padres	78	Clear	Night	0	1	
79	OCT	2	42473	Tuesday	Giants	83	Clear	Night	0	0	



The 8th Tuesday game had the highest attendance, which would be the July 17th game against the Phillies. I don't believe that Phillies vs Dodgers games are a more popular rivalry (not like Mets vs Yankee games) so I am not sure why attendance would be higher for that game with the data that I have. Now I am curious if the Phillies games had higher attendance overall than other games.

In [277...

```
phillies = df[df["opponent"] == "Phillies"]
not_phillies = df[df["opponent"] != "Phillies"]
```

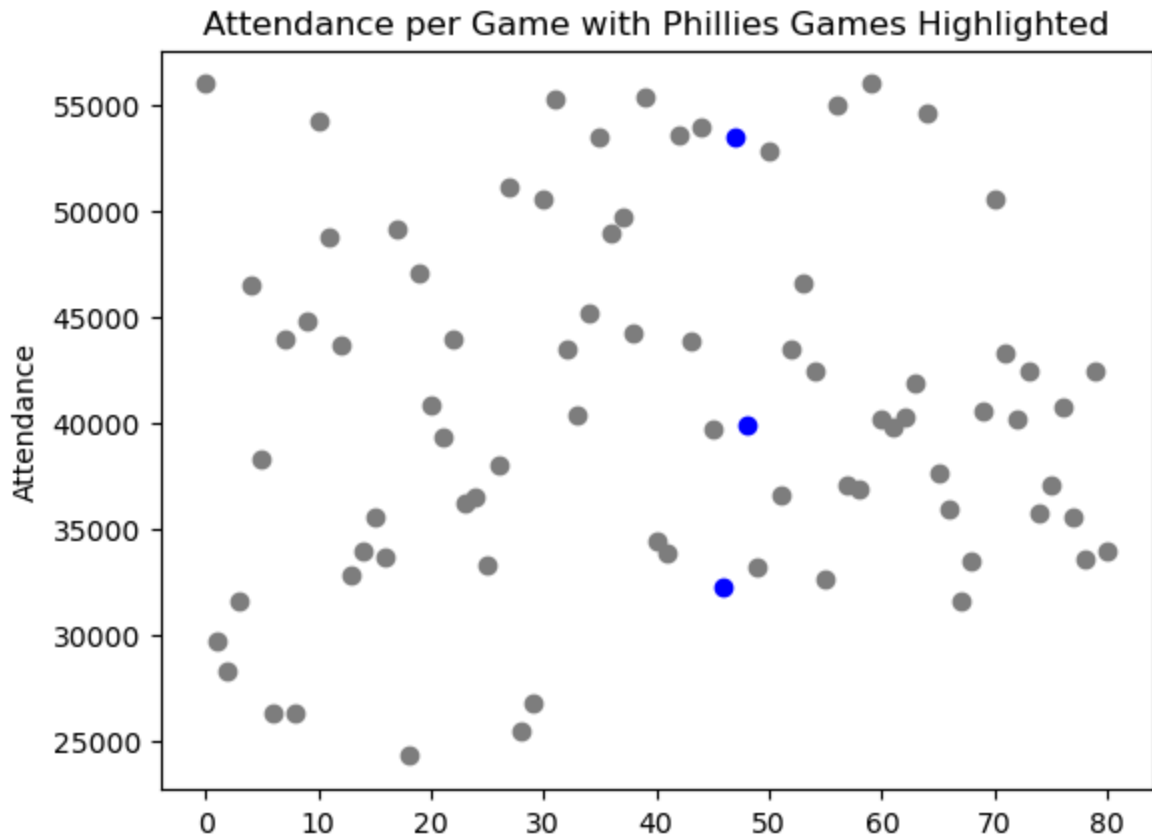
In [281...

```
plt.scatter(not_phillies.index, not_phillies["attend"], color="gray")
plt.scatter(phillies.index, phillies["attend"], color="blue")

plt.title("Attendance per Game with Phillies Games Highlighted")
plt.ylabel("Attendance")
```

Out[281...

```
Text(0, 0.5, 'Attendance')
```



As expected, Phillies games do not have higher attendance overall compared to other games with other opponents.

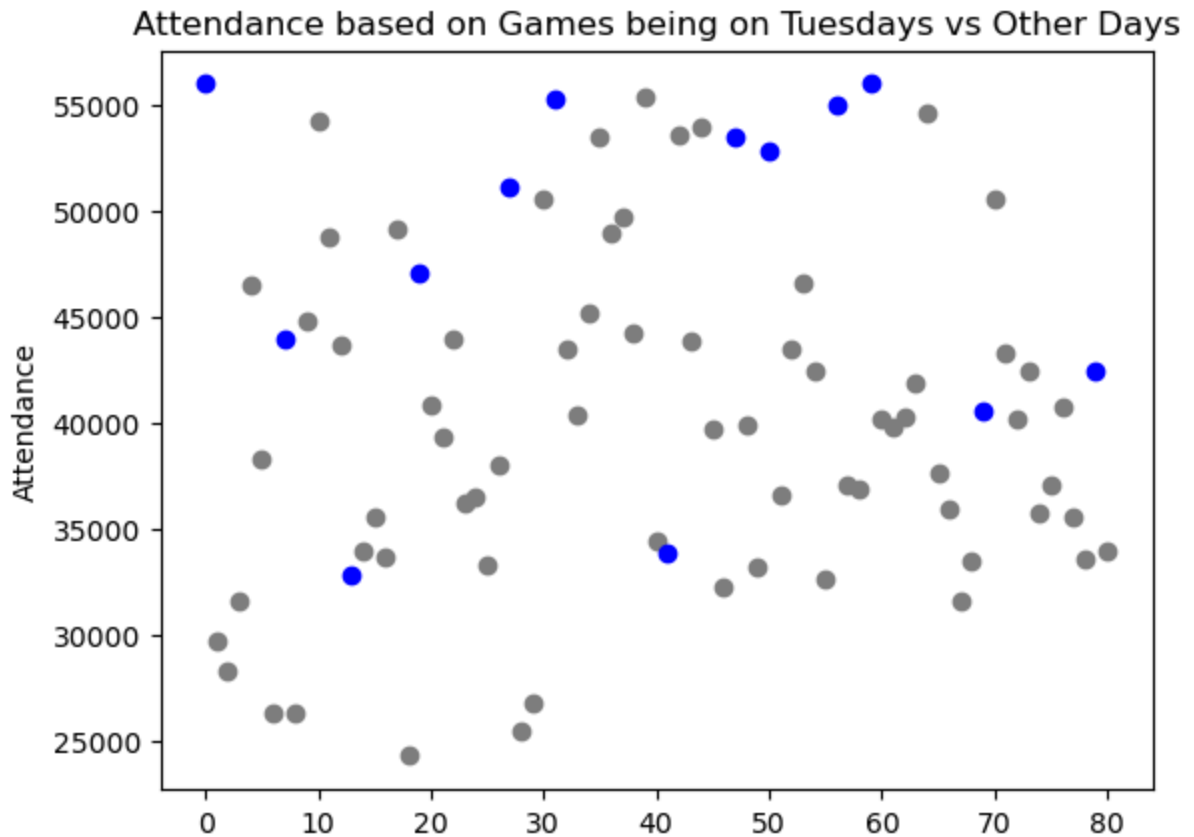
For a final chart, I want to see how Tuesday games compare to all other game days in a scatterplot.

```
In [285... tuesdays = df[df["day_of_week"] == "Tuesday"]
not_tues = df[df["day_of_week"] != "Tuesday"]
```

```
In [289... plt.scatter(not_tues.index, not_tues["attend"], color="gray")
plt.scatter(tuesdays.index, tuesdays["attend"], color="blue")

plt.title("Attendance based on Games being on Tuesdays vs Other Days")
plt.ylabel("Attendance")
```

```
Out[289... Text(0, 0.5, 'Attendance')
```



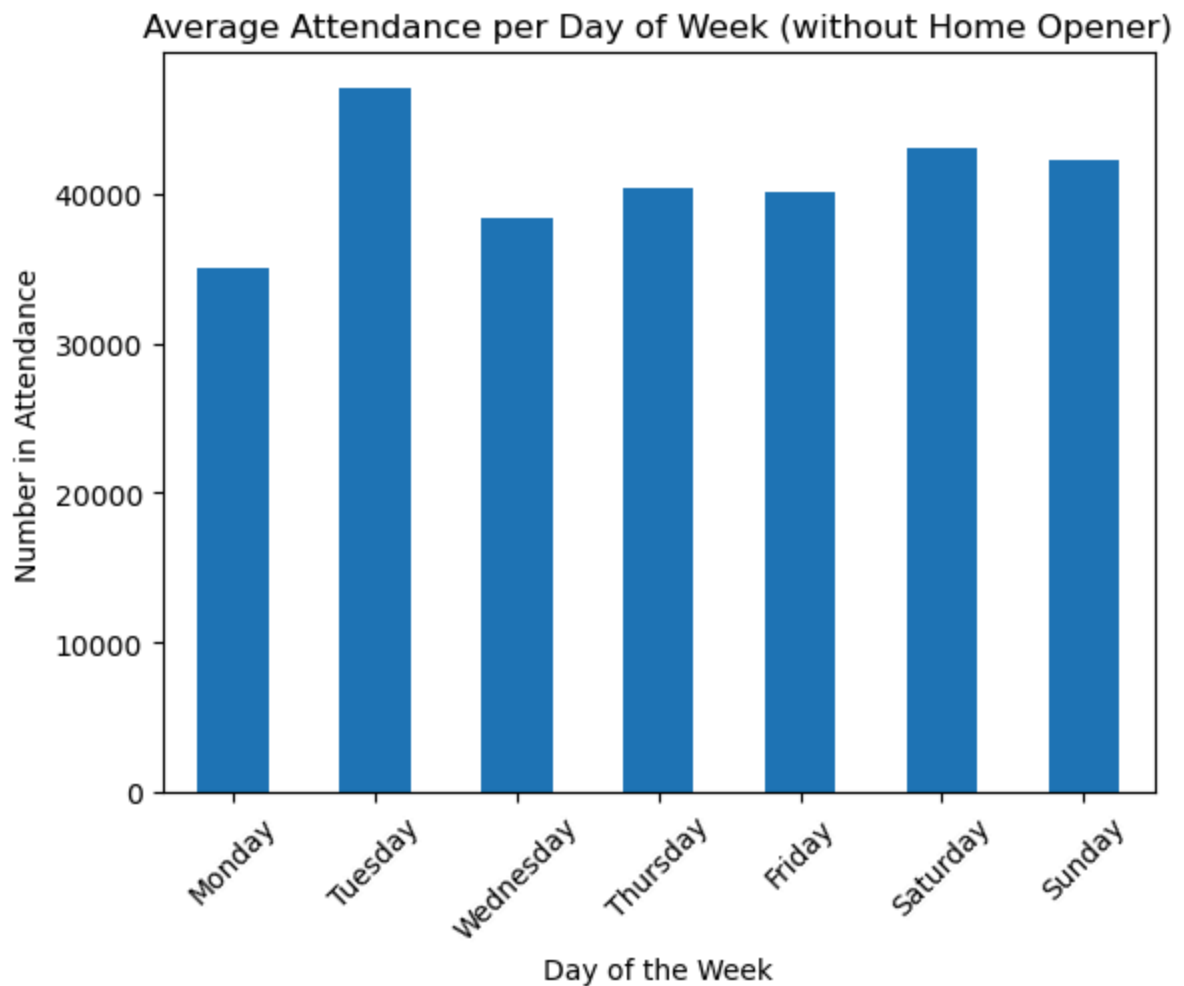
With this scatterplot, I'm led to believe that just because a game is on Tuesday does not make it more popular necessarily than other games, but the games with bobble heads might be causing the increase in attendance.

Finally, I do want to see if Tuesdays had the highest attendance with the home opener data point removed to see if it still has higher attendance compared to other days.

```
In [302... df_without_opener = df.drop(df.index[0])
```

```
In [304... attend_dow_wo_opener = df_without_opener.groupby("day_of_week")["attend"].mean().re
```

```
In [306... attend_dow_wo_opener.plot(kind="bar")
plt.title("Average Attendance per Day of Week (without Home Opener)")
plt.ylabel("Number in Attendance")
plt.xlabel("Day of the Week")
plt.xticks(rotation=45)
plt.show()
```



Conclusion and Recommendation

Based on my analysis, it looks like bobblehead giveaways increase the attendance at baseball games. On average the highest attendance day of the week was Tuesdays, and Tuesdays also had the most bobblehead giveaways. Tuesday games with the bobblehead giveaway had higher attendance on average compared to Tuesday games without the bobblehead giveaway. Regardless of the day of the week, games that included bobblehead giveaways had the highest attendance (not including the home opener day). Overall, I would recommend to add more bobblehead giveaway days to increase attendance.

In []: